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Roll No

MI-402 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Mining Environment - I
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain the various sources of heat and humidity in mines and suggest the measures for minimizing heat and humidity in mines.
2. Explain the following :
 - a) Drilling and Blasting
 - b) Extension of centre line
 - c) Drivages of Incline
3. How “Flow of Air in Ducts and mine Roadways” is maintained? Explain.
4. A fan ventilating a heading through a duct of 600mm diameter circulates $4\text{m}^3/\text{s}$ of air at the face. Calculate the heat added to the air by the fan, if input power of the fan is equal to 2.9kW.
5.
 - a) Explain thermodynamics of natural ventilation.
 - b) Explain different air measuring instruments used in mines.
6. Explain the properties, occurrence and physical effect of mine gases CO_2 and CO .
7. What are the special methods of shaft sinking? Under what circumstances are they followed? State their limitations.

OR

Write “Chezy’s and Atkinson’s Equations” and explain their utility.

8. Write short notes on any two of the following :
 - a) Reynolds Number
 - b) Types of Portable Lamps
 - c) Types of Openings
 - d) Fundamental of Air flow
